

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 5: Fuel Planning & Non-Linear CG Movement

Syllabus Source: CAE Oxford (pp. 89-106) | Pass Target: $\geq 90\%$ AviationExam

1. FUEL TANKS GEOMETRY & CG SHIFT IN FLIGHT

Unlike passenger mass or cargo which remain stationary in flight, fuel mass decreases continuously throughout the trip. Because fuel tanks have complex 3D shapes across swept wings, fuel consumption causes non-linear Center of Gravity movement:

- **Swept-Wing Fuel Tank Arm:** As wings sweep backwards, wingtip tanks are located significantly further AFT than the wing root center tank. Burning fuel from different tanks dramatically moves the aircraft CG forward or aft!
- **Wing Bending Relief & Standard Burn Sequence:** To minimize structural wing root bending moments, commercial jets burn fuel in a strict sequence: CENTER TANK FIRST, WING TANKS LAST! Burning the fuselage center tank first relieves wing-root stress because fuel weight inside the wings counteracts aerodynamic upward wing lift.
- **Unusable Fuel (Trapped Fuel):** Fuel remaining in pipework, pumps, and tank bottoms that cannot be drained into engines during flight. **REGULATION:** Unusable fuel is legally part of BASIC EMPTY MASS (BEM)! It is NEVER counted as usable trip fuel.

2. IN-FLIGHT FUEL JETTISON (FUEL DUMPING)

If a heavy aircraft experiences an emergency immediately after take-off, it may be above its Maximum Structural Landing Mass (MLM):

- **Fuel Jettison System:** Dumps fuel overboard through wingtip nozzles to reduce aircraft mass down to MLM. Jettison valves close automatically to preserve reserve fuel (minimum holding fuel to land).
- **CG Impact of Jettisoning:** Because jettisoning typically empties center tanks and outer wing tanks rapidly, the flight crew must monitor that the CG remains strictly within the certified in-flight envelope during dumping.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Fuel Burnoff & Landing CG Calculation

Problem: TOM = 50,000 kg, Take-Off CG = 18.00 m (Moment = 900,000 kg.m). Trip fuel burned = 8,000 kg at average arm 21.00 m. Step 1: Calculate Landing Mass LM = TOM - Trip Fuel = 50,000 - 8,000 = 42,000 kg. Step 2: Moment of burned fuel = 8,000 kg x 21.00 m = 168,000 kg.m. Step 3: Landing Moment = Take-Off Moment - Burned Moment = 900,000 - 168,000 = 732,000 kg.m. Step 4: Landing CG = 732,000 / 42,000 = 17.428 meters. Rule: Burning fuel aft of CG moves the landing CG forward!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Tracking CG Movement from Take-Off to Landing

SCENARIO (Typical Exam Question): An aircraft departs with: - Take-Off Mass (TOM):

50,000 kg - Take-Off CG: 18.00 meters aft of datum - Total Take-Off Moment = 50,000 x

18.00 = 900,000 kg.m - Trip Fuel to be burned in flight: 8,000 kg - The average Arm of the burned fuel is: 21.00 meters aft of datum (located AFT of the CG!) QUESTION:

Calculate the aircraft's Landing Mass (LM) and the new Landing CG position. SOLUTION

PASO A PASO (FOR DUMMIES): Step 1: Calculate Landing Mass (LM): - LM = TOM - Trip

Fuel = 50,000 kg - 8,000 kg = 42,000 kg. Step 2: Calculate the Moment of the Burned

Fuel: - Fuel Moment = Fuel Mass x Fuel Arm = 8,000 kg x 21.00 m = 168,000

kg.m. Step 3: Subtract the burned fuel moment from the Take-Off Moment: - Landing

Moment = Take-Off Moment - Fuel Moment - Landing Moment = 900,000 kg.m - 168,000

kg.m = 732,000 kg.m. Step 4: Divide Landing Moment by Landing Mass to find Landing

CG: - Landing CG = Landing Moment / Landing Mass - Landing CG = 732,000 kg.m /

42,000 kg = 17.4285 m -> 17.43 meters aft of datum! PHYSICAL EXPLANATION FOR

DUMMIES: Because the burned fuel was located at 21 m (AFT of the 18 m CG), burning it removed weight from the back of the airplane, causing the nose to tilt down and

shifting the CG FORWARD from 18.00 m to 17.43 m!

[!] EXAM TRAP (AVIATIONEXAM) - The Non-Linear Fuel Curve Trap

- Center tanks burn first -> CG typically moves AFT initially. - Wing tanks burn

second -> Swept-wing fuel burn moves CG back FORWARD. - Exam Trap: You CANNOT assume

that if Take-off CG and Landing CG are both within limits, the aircraft was in limits

throughout the entire cruise! You must check the intermediate burn curve peak!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Under EASA AIR-OPS, which category does 'UNUSABLE FUEL' belong to?**

- [A]: Trip Fuel
- [B]: Contingency Fuel
- [C]: BASIC EMPTY MASS (BEM)
- [D]: Traffic Load

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Unusable fuel is the fuel trapped in pipes and tank sumps that cannot reach the engines. It is a permanent fluid load counted as part of the aircraft's

Basic Empty Mass.

Q2: Why is fuel in commercial jet aircraft burned from the center fuselage tank before the wing tanks?

- [A]: Center tank fuel spoils faster
- [B]: Keeping fuel inside the wing tanks as long as possible provides upward inertia and wing bending relief against aerodynamic lift
- [C]: The center pump has higher pressure
- [D]: Wing tanks cannot be pressurized

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Wing fuel weight acts downward along the wingspan, opposing the upward aerodynamic lift, reducing wing-root bending stress. Emptying the center tank first maximizes this structural relief.

Q3: If fuel burned during cruise has an arm located AFT of the current Center of Gravity, what happens to the C

- [A]: The CG moves FORWARD
- [B]: The CG moves rearward
- [C]: The CG stays frozen
- [D]: The CG becomes negative

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Removing mass from behind the pivot point causes the remaining balance to tilt forward, shifting the CG forward.